

GRAPH THEORY – EXAM

8 JANUARY 2026

Please note: This exam consists of 5 problems worth 8 points each, for a total of 40 points. In order to pass this part of the course examination, you will need to obtain at least 18 points out of 40.

For solving this exam, no aids are allowed. If you'd like to use colours, please don't use the colour red.

Unless stated otherwise, all graphs in this exam are finite and simple.

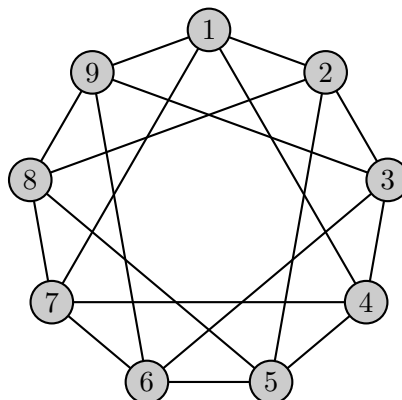
– *Good luck!*

Problem 1. For each of the following statements, decide whether they are true or false, and justify your decision. (2p each)

- (a) A Hamiltonian graph on an even number n of vertices has at least two perfect matchings.
- (b) Any 10-regular graph is 10-colourable.
- (c) For any integers $k \geq 1$, $n \geq 2$, a k -connected graph on n vertices contains at least $\frac{nk}{2}$ edges.
- (d) For every integer $k \geq 1$, there exists an integer n so that every graph on n vertices contains either a clique or an independent vertex set of size at least k (or both).

Problem 2. Consider the following graph G with vertex set

$$V = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$



- (a) We turn G into a weighted graph by giving the edges on the outer cycle the weight 4, and the inner edges the weight 3. Use either Prim's or

Kruskal's algorithm to find a minimum weight spanning tree. Your solution should contain all relevant steps of the algorithm. (4p)

- (b) Use the greedy colouring algorithm with the vertex order

$$1, 2, 3, 4, 5, 6, 7, 8, 9$$

to obtain a proper colouring. (2p)

- (c) Show that $\chi(G) = 3$. (You need to both show that there is a proper colouring with 3 colours, and that there isn't one with fewer colours.) (2p)

Problem 3. a) Define what it means for a graph H to be a minor of a graph G . (Or, if you prefer, a topological minor.) (1p)

- b) State the Wagner-Kuratowski theorem, which characterises planar graphs in terms of their minors and topological minors. (2p)

- c) For which integers a, b, c with $a \geq b \geq c \geq 1$ is the complete 3-partite graph $K_{a,b,c}$ planar? (5p)

Problem 4. Let $n \geq 1$ be an integer and $p \in [0, 1]$.

- (a) Define the random graph $G(n, p)$, and what it means for a graph property (P) to hold 'with high probability'. (2p)

- (b) Let $G \sim G(n, p)$, and for an integer $k \geq 3$, let X_k be the number of cycles of length k in G . Let $X = \sum_{k \geq 3} X_k$ count the number of cycles in the graph $G \sim G(n, p)$. Prove that

$$\mathbb{E}[X] \leq \sum_{k \geq 3} (n^k p^k). \quad (3p)$$

- (c) Use the first moment method to show that if $p \ll \frac{1}{n}$, then with high probability, $G \sim G(n, p)$ contains no cycles. (3p)

Note: You may use without proof that if $f(n) \rightarrow 0$ as $n \rightarrow \infty$, then $\sum_{t \geq 3} f(n)^t \rightarrow 0$ as $n \rightarrow \infty$.

Problem 5. (a) For an integer $n \geq 1$ and a graph H , define $\text{ex}(n, H)$. (1p)

- (b) Let C_5 be the cycle of length 5, and let $n \geq 2$ be even. Prove that

$$\text{ex}(n, C_5) \geq \frac{n^2}{4}. \quad (2p)$$

- (c) For integers $n \geq 1$ and $r \geq 1$, define the Turán graph $T_r(n)$. (1p)

- (d) Assuming that n is divisible by r , how many edges does $T_r(n)$ have? (1p)

- (e) Let H be a graph with chromatic number $\chi(H) = k \geq 3$, and assume that n is divisible by $k - 1$. Prove that

$$\text{ex}(n, H) \geq \frac{k-2}{2(k-1)} n^2. \quad (3p)$$